ELECTROLYZER TECHNOLOGY

ISOTHERMAL ELECTROLYSIS.

>92% SYSTEM EFFICIENCY.

Pulsed waveform architecture eliminates waste heat. No platinum. No iridium. No membranes. Industrial-grade hydrogen production at unprecedented efficiency using commodity materials.

>92%	8×	<30°C	0
SYSTEM EFFICIENCY	LESS HEAT LOSS	OPERATING TEMP	RARE EARTH METALS

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Technical Overview, Milestones, Innovative Green Hydrogen Electrolyzer Design, Certification Plans, and Growth Strategy

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1. INTRODUCTION

Tobe Energy is developing and commercializing next-generation, membrane-less electrolyzers for efficient green hydrogen production. Our systems utilize a novel LLC resonant converter topology to achieve high electrical efficiency (**>92% system efficiency, HHV**) while operating near ambient temperature and eliminating reliance on platinum-group metals or other rare earth materials. This document outlines our current system capabilities, a clear stepwise scale-up plan, intellectual property strategy, manufacturing approach, and product integration pathways. We have successfully advanced from pilot-scale (5–10 kg/day) to commercial pilot-scale (25 kg/day) systems, validating our core technology and de-risking subsequent scaling efforts.

2. TECHNICAL SPECIFICATIONS AND CURRENT PROTOTYPES

2.1 Pilot Scale: 5 kg/day and 10 kg/day Electrolyzers

- **Status:** Fully operational and validated with over **1,000 hours** of cumulative run-time.
- **Key Finding:** Linear scalability from 5 kg/day to 10 kg/day units confirmed with no degradation in efficiency, providing a strong basis for further scaling.
- **Validation:** Prototypes have successfully powered a loaded generator in field tests, demonstrating practical application.

2.2 Commercial Pilot Scale: 25 kg/day Electrolyzer

- **Status:** Achieved. The 25 kg/day system is operational, further validating the technology's performance and design robustness at an increased scale.
- **Significance:** This milestone confirms the system's readiness for initial commercial deployments and provides critical data for subsequent scaling.

2.3 Core Technology and Materials

- **Electrodes:** Standard 304 Stainless Steel (SS304).
- **No Catalyst Required:** Eliminates the need for expensive, supply-chain-constrained platinum-group metals or other rare earth materials common in traditional PEM or AEM electrolyzers.
- **Housing and Components:** Current prototypes utilize 3D-printed parts for rapid iteration. Production units will transition to injection-molded components for optimized cost and manufacturing speed.
- **Electrical Topology:** LLC Resonant Converter Design
  - The electrolyzer cell functions as the primary capacitive load element (the "C" in an LLC resonant circuit).
  - Operation at resonant frequency achieves a high Q-factor (approximately 5), enabling voltage amplification across the cell by a factor of five.
  - Pulsed high-voltage DC at precisely controlled frequencies avoids operation in the infrared (IR) band, significantly reducing thermal losses and enabling near-ambient temperature operation.
- **Temperature Operation:**
  - Maximum observed temperature did not exceed **~30°C** during 1,000+ hours of continuous operation across pilot units.
  - This near-ambient temperature performance is a direct result of minimized resistive (I^2R) heating due to the resonant circuit design.

2.4 Performance Metrics (Validated across 5–25 kg/day Units)

- **Electrical Efficiency:**
  - Verified at **~94.7% stack efficiency (LHV)** under optimized conditions, excluding external lab-grade DC power supply losses.
  - With integrated power electronics (currently in development, targeting >95% efficiency), overall system efficiency is projected to consistently meet or exceed **>92% HHV**.
- **Production Costs:**
  - At **$0.035/kWh** (secured industrial electricity rate), hydrogen production cost is **$1.47/kg**.
  - This is already competitive with, and with optimized power sourcing can be well below, steam-methane-reformed (SMR) hydrogen, which typically costs $1.50–$2.00/kg without carbon capture.

2.5 Scalability Foundations

- Hydrogen production scales linearly with applied power and electrode surface area. This has been consistently observed in the progression from 5 kg/day to 10 kg/day, and subsequently to the 25 kg/day system.
- The modular design approach allows for predictable scaling with minimal re-engineering risk.

3. PRODUCT DEVELOPMENT TIMELINE AND MILESTONES

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Our product roadmap is structured for systematic scaling, continuous technological refinement, and alignment with market-entry requirements, including certifications and intellectual property protection.

TARGET DATE	MILESTONE DESCRIPTION	KEY GOALS & DELIVERABLES
Achieved	Pilot Scale	Stable, repeatable operation confirmed; 5–10 kg/day units fully characterized. Extensive performance data (efficiency, reliability, maintenance intervals) gathered.
Achieved	Commercial Pilot Scale	25 kg/day system confirmed; Bill of Materials (BoM) finalized for standardized production; Full patent application filed (Q1 2025); UL certification process initiated (Q1–Q2 2025).
Q1 2026	Intermediate Scale	125 kg/day system development and validation. Validate system performance, reliability, and manufacturing processes at 5x scale-up from 25 kg/day unit; de-risk transition to industrial volumes; refine modular design for mass production.
Q3 2026	Industrial Scale	Commercial launch of the **250+ kg/day** industrial-scale unit. Optimize supply chain for volume; implement CNC winding and injection molding for cost and lead time reduction.
Q1 2027	Hydrogen-Optimized Applications	Product lines integrated with configurations for key sectors; achieve necessary ISO and CE marks for international market access.
Q3 2027	Integrated Turnkey Packages	Fully deliver integrated packages simplifying hydrogen solutions (on-site production + end-use equipment). Enhance value and control through further vertical integration of manufacturing.

4. FUTURE PLANS & PRODUCT EVOLUTION

4.1 Scale-Up Strategy

Tobe Energy's scale-up strategy focuses on methodical de-risking and validation at each stage, leveraging our modular design and advanced manufacturing techniques.

25 kg/day System (Achieved):

- Successfully validated the LLC resonant design at a commercially relevant pilot scale.
- Robust quality control protocols for power electronics and component manufacturing are being refined based on this system's production and operational data.
- Standardized modular unit design principles have been established for replication.

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125 kg/day System (Target: Q1 2026):

- This intermediate scale serves as a critical de-risking step, validating the system architecture and manufacturing processes before full industrial-scale production.
- Focus on refining the assembly of larger electrode stacks and ensuring consistent performance across scaled-up power electronics.
- Allows for early engagement with industrial partners for pre-commercial trials and feedback.

250+ kg/day System (Target: Q3 2026):

- Scale to industrial volumes primarily by increasing the number of parallel electrode stacks within a standardized module, leveraging learnings from the 125 kg/day system.
- Confirm linear scaling: efficiency and cost-effectiveness must remain consistent or improve.
- Enhance vertical integration: in-house manufacturing of power transformers (CNC-wound), inductors, PCBs, and potentially capacitors to ensure supply chain resilience and cost control.

Beyond 250 kg/day:

- Deploy modular "clusters" of 250 kg/day units sited directly at or near industrial off-takers, minimizing hydrogen transportation costs.
- Achieve a flexible manufacturing model where capacity expansion is accomplished through replication of proven, standardized unit designs.

4.2 Hydrogen-Optimized Applications (Target: Q1 2027 Onward)

Develop integrated solutions pairing the Tobe electrolyzer directly with end-use applications:

- **Power Generators:** Compact, hydrogen-fueled gensets for remote, backup, or off-grid power.
- **Fuel Cells:** Integrated systems for high-efficiency, grid-independent power solutions.

- **Combustion Applications:** Collaborate with partners (e.g., Tulsa Combustion) to provide "drop-in" hydrogen solutions for industrial burners, kilns, and furnaces. Innovations will emphasize "plug-and-play" integration, simplifying end-user adoption of green hydrogen.

4.3 Integrated Turnkey Packages (Target: Q3 2027)

Deliver comprehensive turnkey solutions:

- On-site green hydrogen production unit combined with compatible downstream device (fuel cell, combustion system, or power generator).
- This single-vendor approach will reduce complexity for customers, ensure supply reliability, and streamline maintenance and support.

5. INTELLECTUAL PROPERTY AND CERTIFICATIONS

5.1 Intellectual Property (IP)

- **Provisional Patent Granted (March 27, 2024):** Protects the core capacitive electrolysis design and resonant circuit configuration.
- **Full Patent Application (Filed Q1 2025):** Covers electrolyzer design, proprietary manufacturing methods, and system integration strategies.
- **Ongoing IP Strategy:** Includes trade secrets for specific manufacturing processes (e.g., custom magnetics fabrication, proprietary PCB layouts) and internal software/firmware for Arduino-based control systems. Design patents for integrated module aesthetics and user interfaces are also under consideration.

5.2 Certifications & Standards

- **UL Certification (Process Initiated Q1–Q2 2025):** Essential for product safety compliance in North American markets. Progress is aligned with commercial pilot system validation.
- **ISO & CE Marking (Research & Planning Underway):** CE marking is required for market access in the European Union. Relevant ISO standards (e.g., ISO 22734 for hydrogen generators) will be pursued to ensure international credibility and marketability. Timelines for achieving these certifications align with industrial-scale system availability (post-Q3 2026).
- **Additional certifications** related to hydrogen purity and environmental management systems will be pursued based on target market requirements.

6. DATA COLLECTION, TESTING PROTOCOLS, AND REVISIONS

- **Testing Regimen:** Continuous data logging via Arduino-based control systems during lab tests, field trials, and extended operation. Key metrics include efficiency (electrical and hydrogen output), output purity, load response, and long-term reliability.
- **Thermal Performance:** Thermal imaging is employed, though minimal heat generation (<30°C operation) reduces the criticality of extensive thermal stress testing typical for other electrolyzer types.

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- **Revisions to Technical Specifications:** Specifications are updated dynamically upon validation of performance improvements or design changes. Formal reviews occur annually or following major milestone achievements.
- **Durability & Lifetime Testing:** Focus is on continuous operation under rated loads to assess electrode integrity and power electronics durability. Future accelerated testing may involve stress-testing power electronics at higher loads or varied frequencies to confirm robust design margins.

7. MANUFACTURING AND SUPPLY CHAIN

Current State (Prototype & Pilot Production):

- Transformers and inductors are currently hand-wound for iterative design.
- 3D printing is utilized for rapid prototyping of small parts and enclosures.

Future State (Scaled Production — Phased from Q1 2026):

- **CNC Winding:** Implementation for transformers and inductors planned to coincide with 125 kg/day system development, reducing fabrication time from hours to minutes per component.
- **Injection Molding:** Transition to injection-molded housings and support structures for 125 kg/day and 250+ kg/day systems to enable rapid, cost-effective mass production.
- **Vertical Integration:** In-house manufacturing of critical power electronics, including PCBs, is planned to control costs, ensure quality, and mitigate supply chain risks. Sourcing or in-house production of specialized capacitors is also under evaluation.

Materials Sourcing:

- Reliance on commonly available industrial materials (e.g., SS304) ensures cost-effectiveness and supply chain stability, avoiding dependence on volatile rare earth or PGM markets.

8. CONCLUSION

Tobe Energy's R&D and product roadmap outlines a clear, data-driven progression from achieved pilot and commercial pilot-scale systems (up to **25 kg/day**) towards larger industrial-scale hydrogen production. The successful validation of our high-efficiency resonant circuit technology, combined with the use of readily available materials and a strategic approach to vertical integration and IP protection, positions Tobe Energy to effectively meet the increasing demand for cost-competitive green hydrogen.

The introduction of the **125 kg/day** intermediate system (target Q1 2026) is a key de-risking step towards the **250+ kg/day** industrial units (target Q3 2026). Achievement of these milestones, alongside ongoing testing, timely technical specification updates, and key certifications, will solidify our market position.

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We project that hydrogen-optimized applications will be available by **Q1 2027**, followed by fully integrated turnkey solutions by **Q3 2027**, significantly accelerating the adoption of affordable, emissions-free hydrogen across multiple industries.

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